Listing 8.5 `document_query_chromadb.py` (relevant code sections)

```
embeddings = [get_embedding(doc) for doc in documents]     #1
ids = [f"id{i}" for i in range(len(documents))]           #1

chroma_client = chromadb.Client()               #2
collection = chroma_client.create_collection(
                        name="documents")       #2
collection.add(     #3
    embeddings=embeddings,
    documents=documents,
    ids=ids
)

def query_chromadb(query, top_n=2):      #4
    query_embedding = get_embedding(query)
    results = collection.query(
        query_embeddings=[query_embedding],
        n_results=top_n
    )
    return [(id, score, text) for id, score, text in
            zip(results['ids'][0],
                results['distances'][0],
                results['documents'][0])]

while True:     #5
    query = input("Enter a search query (or 'exit' to stop): ")
    if query.lower() == 'exit':
        break
    top_n = int(input("How many top matches do you want to see? "))
    search_results = query_chromadb(query, top_n)

    print("Top Matched Documents:")
    for id, score, text in search_results:
        print(f"""
ID:{id} TEXT: {text} SCORE: {round(score, 2)}
""")    #5

    print("\n")
###Output
Enter a search query (or 'exit' to stop): dogs are lazy
How many top matches do you want to see? 3
Top Matched Documents:
ID:id7 TEXT: The dog is lazy but the brown fox is quick! SCORE: 0.24
ID:id5 TEXT: The brown fox is quick and the blue dog is lazy! SCORE: 0.28
ID:id2 TEXT: The quick brown fox jumps over the lazy dog. SCORE: 0.29
```

#1 Generates embeddings for each document and assigns an ID
#2 Creates a Chroma DB client and a collection
#3 Adds document embeddings to the collection
#4 Queries the datastore and returns the top n relevant documents
#5 The input loop for user input and output of relevant documents/scores

As the earlier scenario demonstrated, you can now query the documents using semantic meaning rather than just key terms or phrases. These scenarios should now provide the background to see how the retrieval